Project Title: An Empirical Study on an AI-Driven System for Intraday Stock Market Prediction

A Project Report
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Submitted by

Deepesh Sharma
(Roll No:25scs1003000671)

Under the supervision of

Mr. Shobhit Agarwal
Assistant Professor

IILM University, Greater Noida, Uttar Pradesh
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Abstract

This project proposes an automated system for intraday trading that utilizes Artificial Intelligence (AI) to neutralize the risks inherent in manual trading, such as psychological bias and execution errors. Financial markets are defined by their chaotic nature and non-linear volatility, rendering conventional linear forecasting tools inadequate. The primary objective of this system is to enhance profitability by synthesizing historical market data with real-time trend analysis. The core methodology involves an empirical comparison between ARIMA (Auto-Regressive Integrated Moving Average), a standard time-series statistical model, and Artificial Neural Networks (ANN), a machine learning approach capable of complex pattern recognition, as detailed in the foundational study by Venkatarathnam et al.. Additionally, the project incorporates Big Data principles by processing unstructured text from financial news and social platforms to gauge market sentiment. By integrating these technologies, the system aims to pinpoint precise entry and exit targets, thereby maximizing trade efficiency and minimizing exposure for intraday traders.

Keywords: Artificial Neural Networks (ANN), ARIMA, Stock Market Prediction, Sentiment Analysis, Algorithmic Trading.

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1. Introduction

1.1 Overview

Financial ecosystems are intricate and highly dynamic, with asset prices fluctuating due to a myriad of variables ranging from past performance metrics to macroeconomic shifts and public perception. In the context of intraday trading, speed is critical; a latency of mere seconds can transform a potential profit into a significant loss. To navigate these challenges, integrating Artificial Intelligence (AI) and Machine Learning (ML) offers a robust, data-centric mechanism to automate and optimize trading decisions.

1.2 Objectives

- To engineer an automated trading bot capable of parsing historical stock datasets, specifically Open, High, Low, Close, and Volume (OHLCV) metrics.
- To implement and empirically evaluate the forecasting accuracy of ARIMA versus ANN models to identify the superior approach for price prediction.
- To embed Sentiment Analysis capabilities that process textual data from news feeds and social media, capturing real-time market direction.
- To eliminate emotional interference and manual errors from the trading execution process.

1.3 Problem Statement

Manual trading within a single day is highly susceptible to human fallibility and stress, often resulting in missed profitable setups or inconsistent returns. Conventional forecasting techniques, which often rely on simple linear statistics, struggle to decode the complex, non-linear relationships present in massive financial datasets. Moreover, purely numerical models frequently overlook the immediate impact of breaking news and social sentiment, which are crucial drivers of sudden market volatility.

2. Literature Review

Numerous academic works have investigated the utility of AI in finance. The following table synthesizes the core research papers that form the theoretical basis of this project:

S.No	Author(s) & Year	Methodology / Focus	Key Findings / Understanding
1	Venkatarathnam et al. (2024) (Base Paper)	Conducted a comparative analysis of ARIMA and ANN using R programming, specifically on Adani Ports stock data.	ANN excels at detecting complex, non-linear data patterns, whereas ARIMA is highly effective for short-horizon trend forecasting based on past sequences. Hybrid models like GA-ANN are recommended for future optimization.
2	Sizan et al. (2023)	Assessed the performance of Random Forest, Gradient Boosting, and Logistic Regression within the US financial market.	Although Logistic Regression showed high raw accuracy (81.4%), Random Forest delivered the superior F1-score, establishing it as the most balanced and dependable model for managing risk.
3	Sarisa et al. (2024)	Investigated market trends through the lens of Big Data Integration and Sentiment Analysis	Highlighted that integrating unstructured textual data (news, social blogs) with numerical data is essential for gaining a competitive edge and predicting trends in real-time.

S.No	Author(s) & Year	Methodology / Focus	Key Findings / Understanding
4	Chopra & Sharma (2021)	A systematic review covering 148 distinct studies regarding AI in stock forecasting.	Pinpointed critical gaps in current research, specifically regarding input data selection and data pre-processing. Recommended that future work focus on feature selection algorithms.
5	Ferreira et al. (2021)	A broad review of 2,326 papers focusing on AI applications in trading, including portfolio optimization.	Validated that hybrid approaches—those combining multiple AI techniques—consistently outperform singular models in simulated trading environments.

3. Metho dology

3.1 ArchitectureOverview

The proposed system adopts a structured data mining architecture. It ingests two distinct types of input: Historical Data(quantitative) and Textual Data (qualitative/sentiment). These inputs are processed through parallel analytical streams:

- Time-Series Analysis: Utilizes the ARIMA model to identify linear trends and stationarity.
- Pattern Recognition: Deploys ANN and ensemble methods (such as Random Forest) to uncover intricate, non-linear market behaviors.

3.2 Data Framework

Drawing from the empirical framework established by Venkatarathnam et al., the data structure includes:

- Fundamental Ratios: Indicators such as P/E Ratio and Earnings Per Share (EPS).
- Technical Indicators: Tools including Moving Averages, Relative Strength Index (RSI), and Bollinger Bands.
- Sentiment Data: Information scraped from financial news outlets and social media platforms to gauge public mood.

3.3 Software Tools

- Language: Python and R (specifically for statistical modeling tasks).
- Libraries: TensorFlow/Keras (for constructing ANNs), Statsmodels (for ARIMA calculations), and
- Scikit-learn (for Random Forest implementation).
- Data Sources: Yahoo Finance API for numerical data and Twitter API or news feeds for textual sentiment data.

4. Implementation

The execution of the project follows these defined stages:

1. Data Acquisition: Automated retrieval of OHLCV (Open, High, Low, Close, Volume) data sets.
2. Pre-processing: Rectifying missing values, normalizing datasets (Scaling between 0-1), and computing necessary technical indicators.
3. Model Training:
    - The ARIMA model will undergo tuning to identify the ideal (p, d, q) parameters ensuring data stationarity.
    - The ANN model will be trained using hidden layers to recognize historical price patterns.
4. Back-testing: The system will be rigorously tested against historical data to simulate trading outcomes and quantify potential Profit/Loss (P&L).

5. Results and Discussion

It is anticipated that the Artificial Neural Network (ANN) will demonstrate superior performance in volatile market scenarios due to its capacity to learn non-linear correlations. Conversely, ARIMA is expected to offer stability during periods of low market volatility. The addition of Sentiment Analysis is projected to enhance the system's responsiveness to sudden news events, minimizing the lag often associated with purely technical models. The ultimate output will be a clear trading signal (BUY, SELL, HOLD) accompanied by a confidence metric.

6. Conclusion and Future Scope

6.1 Conclusion

Incorporating AI and ML into stock market forecasting dramatically improves decision-making accuracy compared to traditional manual approaches. By merging the statistical rigor of ARIMA with the advanced pattern recognition of ANN, this system offers a powerful mechanism for intraday trading.

6.2 Future Scope

- Hybrid Optimization: Integrating Genetic Algorithms (GA) to automatically tune ANN parameters for better efficiency.
- Reinforcement Learning: Developing agents that learn trading strategies through a reward-penalty system based on profit and loss.
- Live Deployment: Interfacing the bot with a live brokerage API to execute real-time trades.

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